Temporal decoupling of CTLA-4 and ERK sensitivity in brain tumour signaling: a theoretical basis for sequencing checkpoint and MAPK inhibition

Table S3: Immunotherapy Parameters

Table 1: Immunotherapy approach: Targeted entities and drugs.

S.No	Drug	Targets	Kinetic parameter value
1	Ipilimumab	CTLA-4	$k^{+}_{\text{CTLA-4,MDX-CTLA-4}} = 3.83 \times 10^{5}$
2	Ulixertinib	ERK	$k^{+}_{\text{ERK,BVD-523}} = 3.35 \times 10^{7}$
3	Alvocidib	MDM2	$k^{+}_{\text{MDM2,AMG-232}} = 1.2 \times 10^{6}$
4	Lapatinib	RTKs	$k^{+}_{\text{RTKs,GW572016}} = 4.4 \times 10^{6}$

Table 2: Summary of drugs information intervened in treatment state.

Drug	Doses	Start time	Amount	Rate (k)	Interval	Repeated count
Ipilimumab (MDX-CTLA-4)	Dose-1	0	3.0×10^{-9}	0.01	1000	3
	Dose-2	380	1.75×10^{-8}	0.01	1000	3
	Dose-3	760	3.0×10^{-8}	0.01	1000	3
Ulixertinib (BVD-523)	Dose-1	0	1.12×10^{-8}	0.01	1000	3
	Dose-2	380	2.24×10^{-8}	0.01	1000	3
	Dose-3	760	1.12×10^{-7}	0.01	1000	3
Alvocidib (AMG-232)	Dose-1	0	2.92×10^{-8}	0.01	1000	3
	Dose-2	380	5.84×10^{-8}	0.01	1000	3
	Dose-3	760	2.92×10^{-7}	0.01	1000	3
Lapatinib (GW572016)	Dose-1	0	1.9×10^{-7}	0.01	1000	3
	Dose-2	380	3.86×10^{-7}	0.01	1000	3
	Dose-3	760	1.9×10^{-6}	0.01	1000	3